NLP Interview Questions & Answers

Basic to Intermediate | Beginner Friendly

github.com/mehvishsheikh31/NLP

Section 1: NLP Fundamentals

Q1. What is NLP (Natural Language Processing)?

Ans: NLP is a branch of AI that enables computers to understand, interpret, and generate human language. It bridges the gap between human communication and machine understanding.

Examples: spam detection, chatbots, Google Translate, Siri, autocorrect.

Q2. What is the difference between NLU and NLG?

Ans: NLU (Natural Language Understanding) focuses on comprehending the meaning of text (e.g., classifying intent). NLG (Natural Language Generation) focuses on producing human-readable text (e.g., writing summaries). NLP = NLU + NLG together.

Q3. What is a corpus in NLP?

Ans: A corpus is a large structured collection of text used for training or evaluating NLP models. Example: Wikipedia dump, IMDb movie reviews dataset, or Twitter data.

Q4. What is tokenization? Name its types.

Ans: Tokenization is splitting raw text into smaller units called tokens.

- Word tokenization: 'I love NLP' → ['I', 'love', 'NLP']
- Sentence tokenization: splits paragraph into sentences
- Subword tokenization: 'playing' → ['play', '##ing'] (used in BERT)
- Character tokenization: 'cat' → ['c','a','t']

Q5. What are stopwords? Give examples.

Ans: Stopwords are very common words that carry little meaningful information and are often removed before processing. Examples: 'is', 'the', 'a', 'in', 'and', 'of'. However, in some tasks like sentiment analysis, stopwords like 'not' matter and should be kept.

Section 2: Text Preprocessing

Q6. What is stemming? Give an example.

Ans: Stemming reduces a word to its root form by chopping off suffixes — the result may not be a real word. It is fast but less accurate.

- 'running' → 'run' | 'studies' → 'studi' | 'flies' → 'fli'

Algorithm used: Porter Stemmer, Snowball Stemmer.

Q7. What is lemmatization? How is it different from stemming?

Ans: Lemmatization reduces a word to its base dictionary form (lemma) using vocabulary and grammar. It always returns a real word.

Word	Stem	Lemma
running	`run`	`run`
studies	`studi`	`study`
better	`better`	`good`
flies	`fli`	`fly`

Q8. What is the difference between stemming and lemmatization? When to use which?

Ans: Use stemming when speed matters and approximate roots are fine (e.g., search engines). Use lemmatization when accuracy matters and real word forms are needed (e.g., chatbots, Q&A systems).

Q9. What is POS tagging?

Ans: POS (Part-of-Speech) tagging assigns grammatical labels to each word — noun, verb, adjective, etc.

- 'The quick brown fox jumps' → [Det, Adj, Adj, Noun, Verb]

Used in: named entity recognition, parsing, machine translation.

Section 3: Text Representation

Q10. What is Bag of Words (BoW)?

Ans: BoW represents text as a vector of word counts, ignoring order and grammar. Each column is a unique word; each row is a document.

- 'I love cats' and 'cats love me' → same BoW (order ignored)
- Limitation: does not capture word meaning or order

Q11. What is TF-IDF? Why is it better than BoW?

Ans: TF-IDF = Term Frequency x Inverse Document Frequency.

- TF = how often a word appears in a document
- IDF = log(total docs / docs containing word) — penalizes common words
- Result: common words like 'the' get low scores; rare informative words get high scores

Better than BoW because it weights words by importance, not just frequency.

Q12. What is the difference between BoW and TF-IDF?

Feature	Bag of Words	TF-IDF
Scoring	Raw word counts	Weighted by importance
Common words	High score	Low score (penalised)
Speed	Fast	Slightly slower
Use case	Simple baseline	Spam filter, search

Q13. What are word embeddings? Why are they better than BoW?

Ans: Word embeddings are dense vector representations of words where similar words have similar vectors. Unlike BoW which gives sparse, unrelated vectors, embeddings capture semantic meaning.

- 'king' and 'queen' are close in vector space
- 'king' - 'man' + 'woman' ≈ 'queen' (famous Word2Vec analogy)
- Examples: Word2Vec, GloVe, FastText

Q14. What is Word2Vec? What are its two architectures?

Ans: Word2Vec is a neural network model that learns word embeddings from large text. It has two architectures:

- CBOW (Continuous Bag of Words): predicts a word from surrounding context words. Faster, good for frequent words.
- Skip-gram: predicts surrounding context words from a given word. Better for rare words.

Q15. What is cosine similarity? How is it used in NLP?

Ans: Cosine similarity measures the angle between two vectors. Value ranges from -1 (opposite) to 1 (identical).

```
similarity = dot(A, B) / (||A|| * ||B||)
```

Used to find similar words, sentences, or documents. Example: find synonyms using Word2Vec embeddings.

Section 4: Deep Learning for NLP

Q16. What is an RNN? What is its main problem?

Ans: RNN (Recurrent Neural Network) processes text word by word, maintaining a hidden state that carries information from previous steps. Main problem: vanishing gradient — it forgets information from far-back words, making it bad for long sequences.

Q17. What is LSTM and how does it solve the vanishing gradient problem?

Ans: LSTM (Long Short-Term Memory) uses three gates to control information flow:

- Forget gate: decides what old information to discard
- Input gate: decides what new information to store
- Output gate: decides what to output as hidden state

These gates allow LSTM to carry relevant information across long sequences, solving the vanishing gradient problem.

Q18. What is the Embedding layer in Keras/TensorFlow?

Ans: The Embedding layer converts integer word indices into dense vectors. It acts as a lookup table — each word gets a learnable vector of fixed size (embed_dim).

```
Embedding(vocab_size=5000, embed_dim=32, input_length=100)
```

It is the first layer in most deep NLP models, placed before LSTM or other layers.

Q19. What is padding in NLP? Why is it needed?

Ans: Neural networks require fixed-length inputs, but sentences have variable lengths. Padding adds zeros to shorter sequences to make them equal length.

- Pre-padding (default): zeros added at the beginning
- Post-padding: zeros added at the end

```
pad_sequences(sequences, maxlen=100, padding='post')
```

Q20. What is the difference between sigmoid and softmax?

	Sigmoid	Softmax
Output	Single value 0–1	N values summing to 1
Use for	Binary classification	Multiclass (3+ labels)
Loss	binary_crossentropy	categorical_crossentropy
Example	Spam or not spam	Positive / Neutral / Negative

Q21. What is Dropout and why is it used?

Ans: Dropout randomly deactivates a fraction of neurons during training (e.g., Dropout(0.5) = 50% neurons turned off). This prevents overfitting by forcing the network to not rely on any single neuron. It is removed during inference/testing.

Section 5: Transformers & BERT

Q22. What is the attention mechanism in Transformers?

Ans: Attention allows the model to focus on relevant words when processing each word. Unlike RNNs, it looks at ALL words simultaneously rather than one at a time. Key formula:

```
Attention(Q, K, V) = softmax(QK^T / sqrt(d_k)) * V
```

Q = Query, K = Key, V = Value. Each word computes how much attention to pay to every other word.

Q23. What is BERT? What makes it different from older models?

Ans: BERT (Bidirectional Encoder Representations from Transformers) is a pretrained transformer model. Key differences:

- Bidirectional: reads text left-to-right AND right-to-left simultaneously (older models were one direction)
- Pretrained on massive data (Wikipedia + BookCorpus), fine-tuned for specific tasks
- Uses WordPiece tokenization (subwords)
- State-of-the-art on many NLP benchmarks

Q24. What is transfer learning in NLP?

Ans: Transfer learning means taking a model pretrained on a large dataset and fine-tuning it on a smaller task-specific dataset. Instead of training from scratch, we reuse learned representations.

- Pretraining: BERT trained on Wikipedia (Anthropic, Google, etc. spend millions doing this)
- Fine-tuning: you train BERT on your 1000-sample dataset for 3–5 epochs

Benefit: great performance with very little data and compute.

Q25. What is the difference between BERT and GPT?

	BERT	GPT
Direction	Bidirectional	Left-to-right only

Architecture	Encoder only	Decoder only
Best for	Classification, NER, QA	Text generation
Pretraining	Masked language model	Next word prediction

Q26. What is Named Entity Recognition (NER)?

Ans: NER identifies and classifies named entities in text into categories like Person, Organization, Location, Date, etc.

- Input: 'Elon Musk founded SpaceX in California in 2002.'
- Output: Elon Musk [PERSON], SpaceX [ORG], California [LOC], 2002 [DATE]

Used in: information extraction, knowledge graphs, search engines.

Q27. What is the Hugging Face pipeline? Name 4 tasks it supports.

Ans: Hugging Face pipeline() is a high-level API that loads a pretrained model and runs inference in 2–3 lines of code.

- sentiment-analysis: positive/negative classification
- ner: named entity recognition
- question-answering: answer questions from a context paragraph
- fill-mask: predict masked/missing words in a sentence
- text-generation: generate new text from a prompt
- summarization: shorten long text

Section 6: Bonus — Quick Conceptual Questions

Q28. What is the difference between syntax and semantics in NLP?

Ans: Syntax refers to the grammatical structure of sentences (how words are arranged). Semantics refers to the meaning of words and sentences. 'Colorless green ideas sleep furiously' is syntactically correct but semantically nonsensical.

Q29. What is an N-gram? Give an example.

Ans: An N-gram is a contiguous sequence of N words from a text.

- Unigram (N=1): ['I', 'love', 'NLP']
- Bigram (N=2): ['I love', 'love NLP']
- Trigram (N=3): ['I love NLP']

Used in: language models, spell checkers, text prediction.

Q30. What is text classification? Give two real-world examples.

Ans: Text classification assigns a predefined label/category to a piece of text.

- Spam detection: classify email as spam or not spam (binary)
- Sentiment analysis: classify review as positive, neutral, or negative (multiclass)
- Topic classification: classify news articles into sports, politics, technology, etc.

Q31. What evaluation metrics are used in NLP classification tasks?

Ans: Common metrics:

- Accuracy: (correct predictions) / (total predictions) — good for balanced datasets

- Precision: of all predicted positives, how many are actually positive
- Recall: of all actual positives, how many did we correctly predict
- F1-Score: harmonic mean of Precision and Recall — best for imbalanced datasets

Q32. What is the difference between Word2Vec and GloVe?

Ans: Both produce word embeddings but use different approaches:

- Word2Vec: learns embeddings by predicting context words locally (window-based neural network)
- GloVe (Global Vectors): learns from the global word co-occurrence matrix across the entire corpus

In practice, both give similar quality embeddings. GloVe is often faster to train at scale.